



Georgia Tech
College of
Engineering

COE 3001

MECHANICS OF DEFORMABLE BODIES

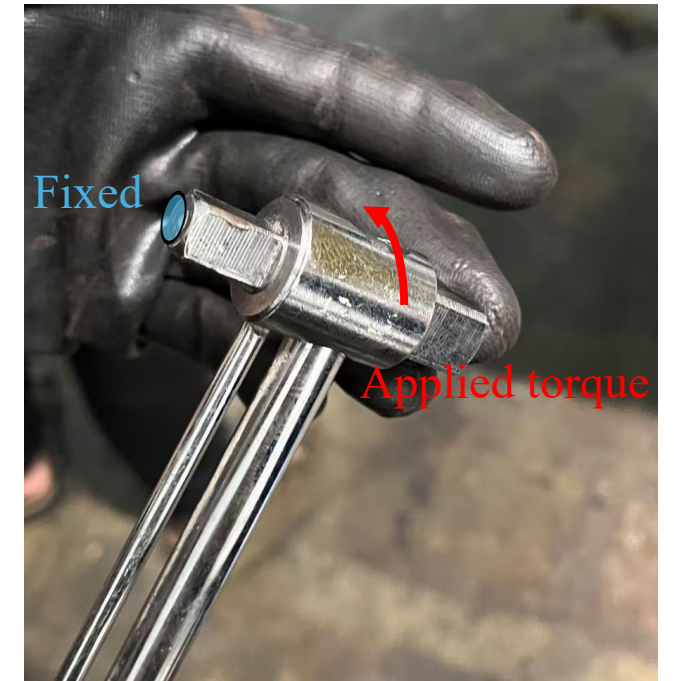
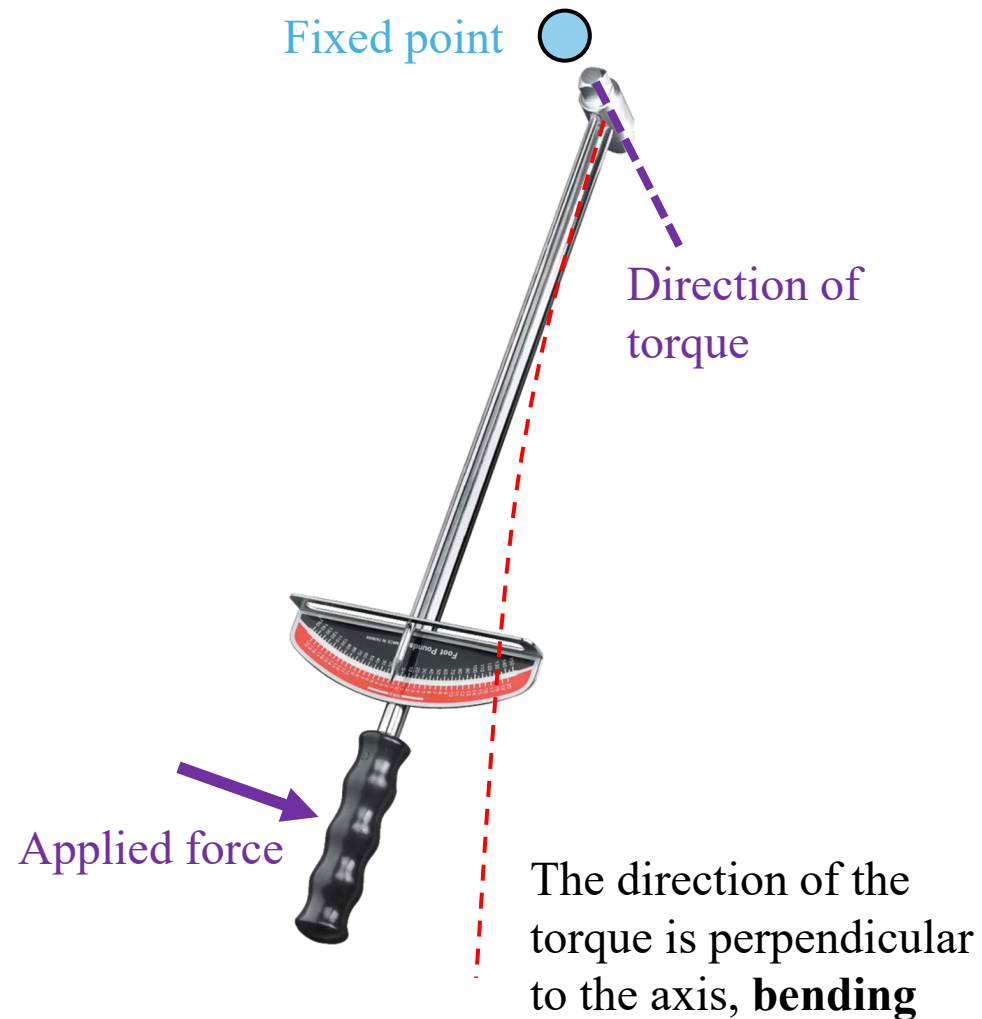
Lecture 11 – Bending: Shear force and bending moment diagrams

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Georgia Institute of Technology

Feb. 23, 2026

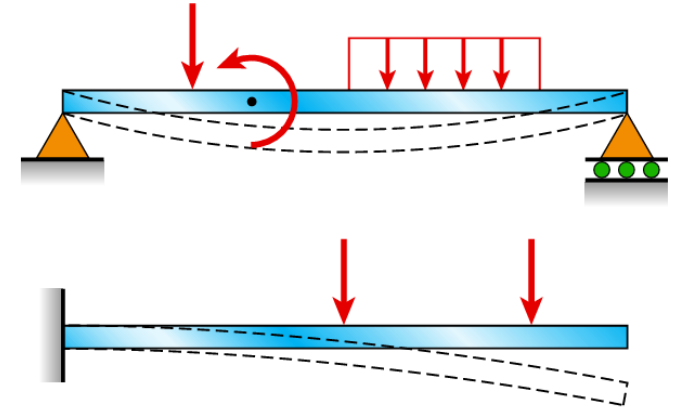
Example: analysis of torque wrench



The direction of the torque is along the axis, **torsion**

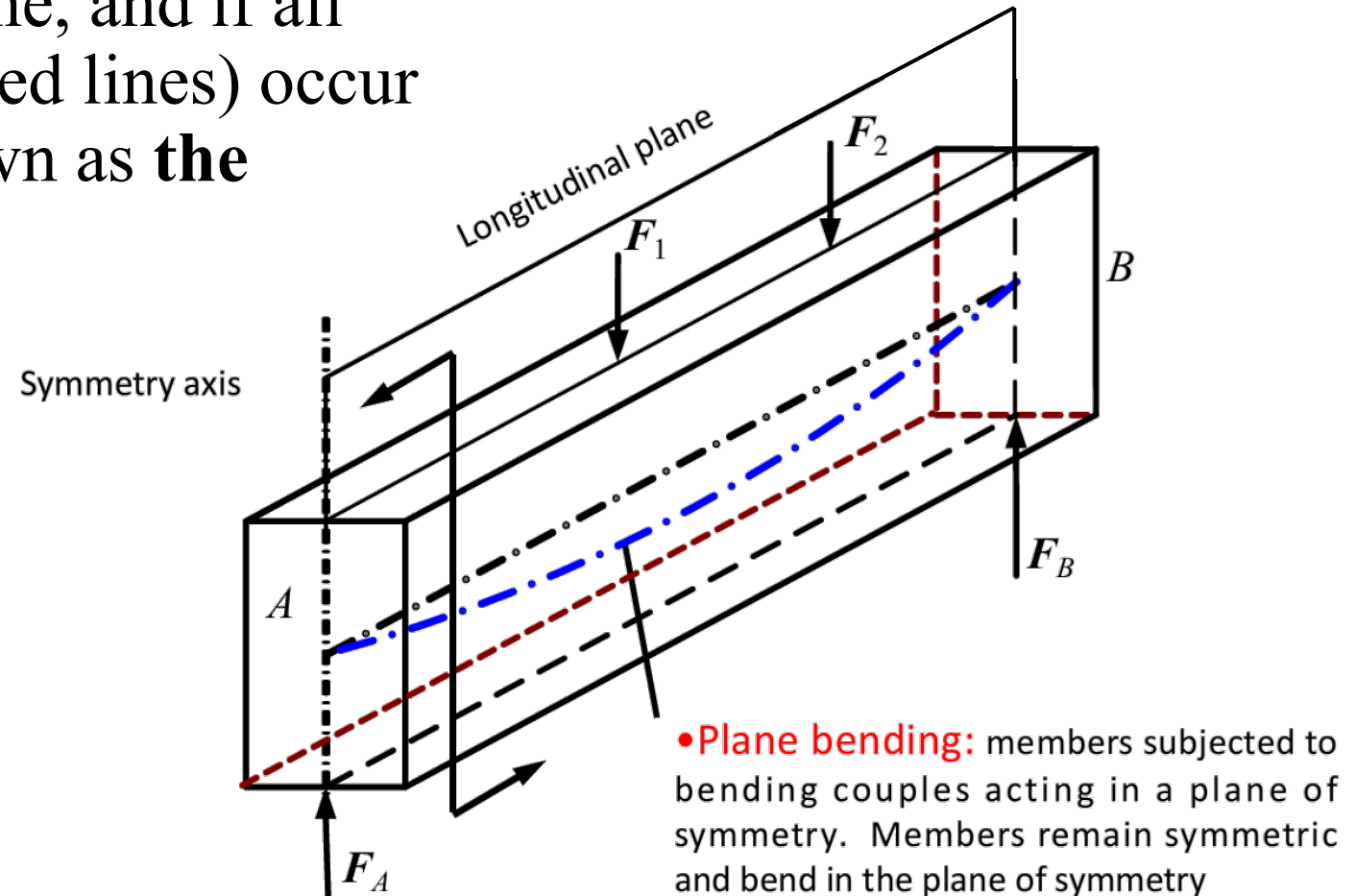
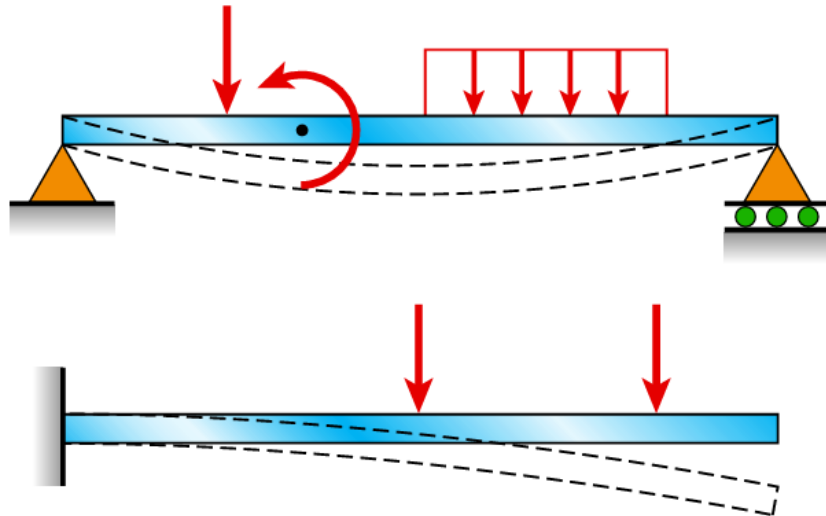
Beams

Structural members subjected to lateral loads, that is, forces or moments having their vectors perpendicular to the axis of the bar.

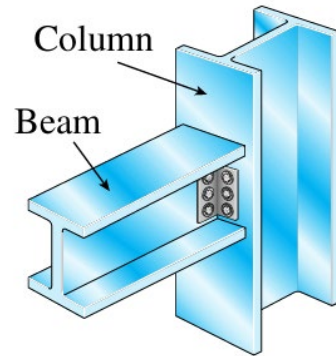


Plane bending (or symmetric bending)

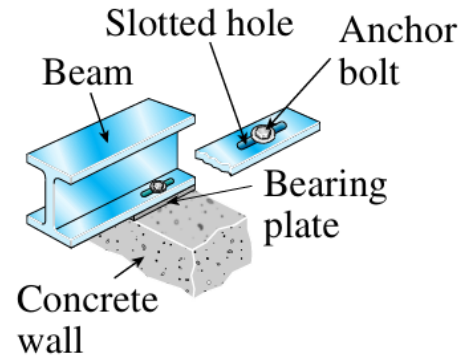
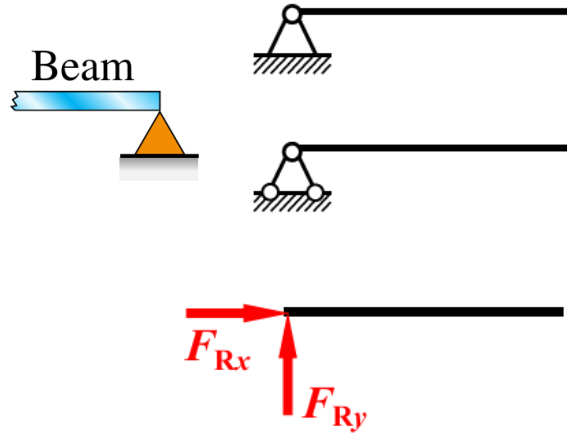
If all loads act in that same plane, and if all deflections (shown by the dashed lines) occur in that plane, that plane is known as **the plane of bending**.



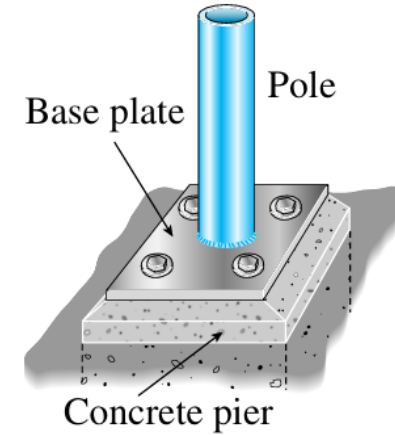
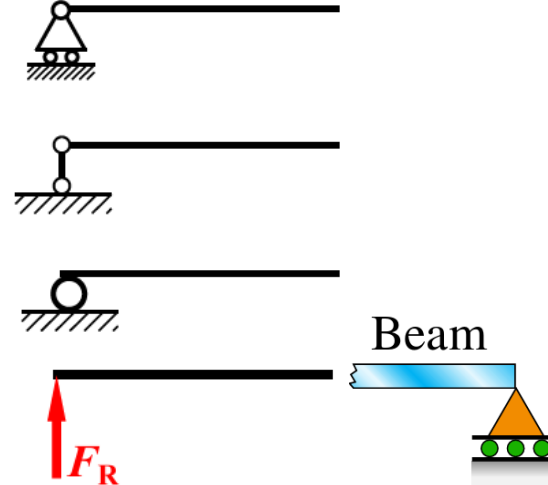
Support types



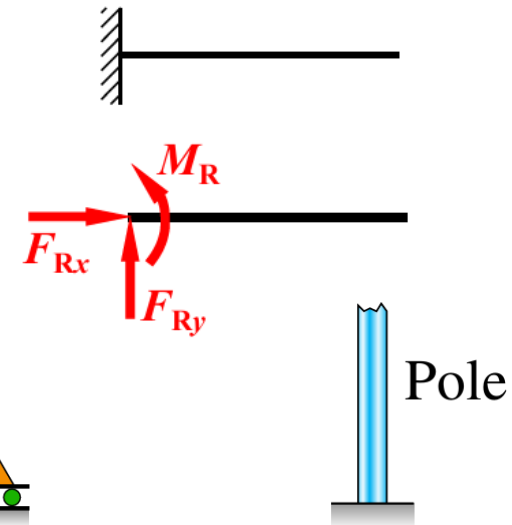
Fixed hinge support



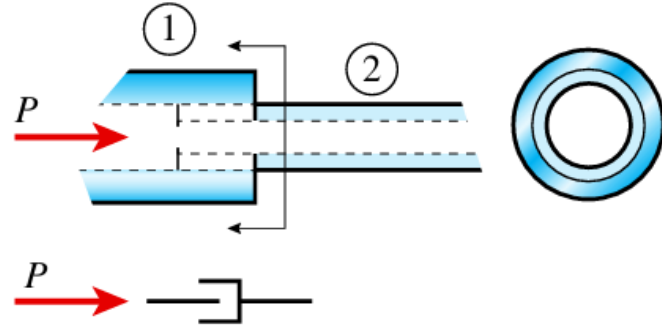
Movable hinge support



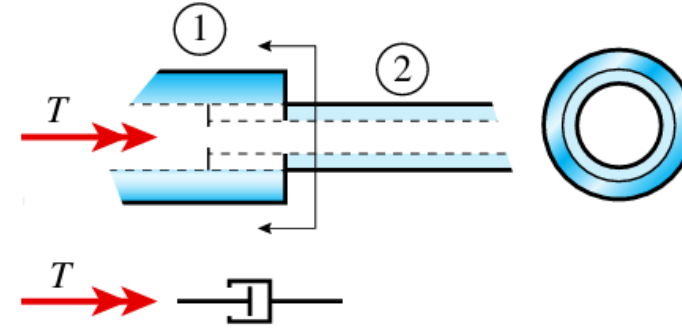
Fixed end



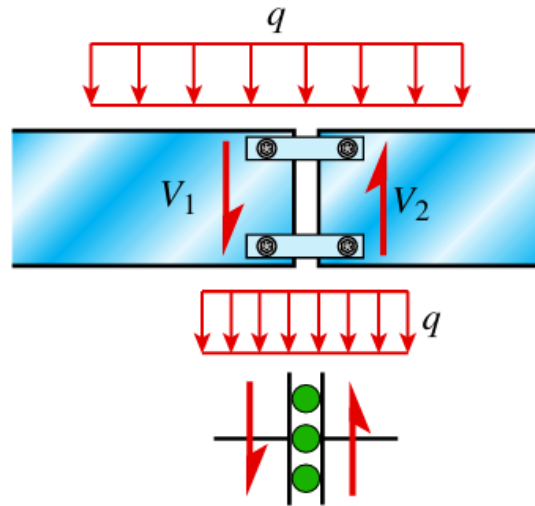
Internal releases



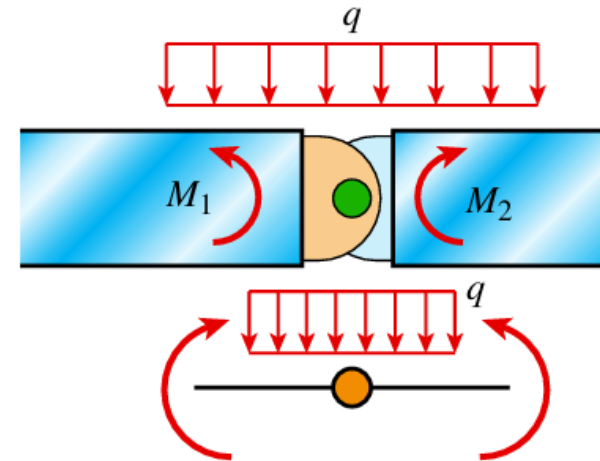
(a) Axial release ($N = 0$; $V, M \neq 0$)



(b) Torque release ($T = 0$; $N, V, M \neq 0$)

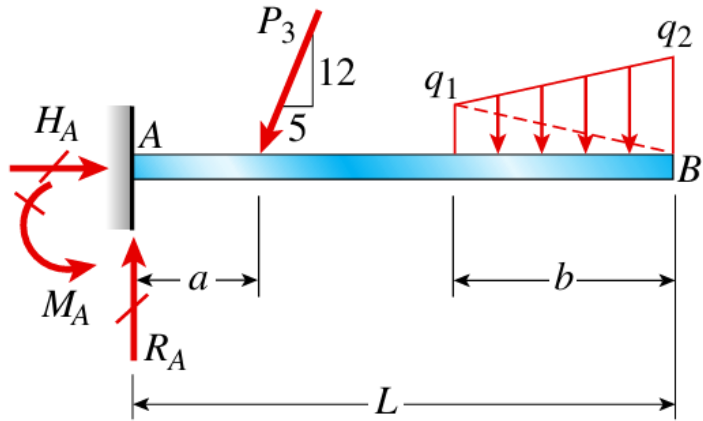


(c) Shear release ($V = 0$; $N, M \neq 0$) ★★



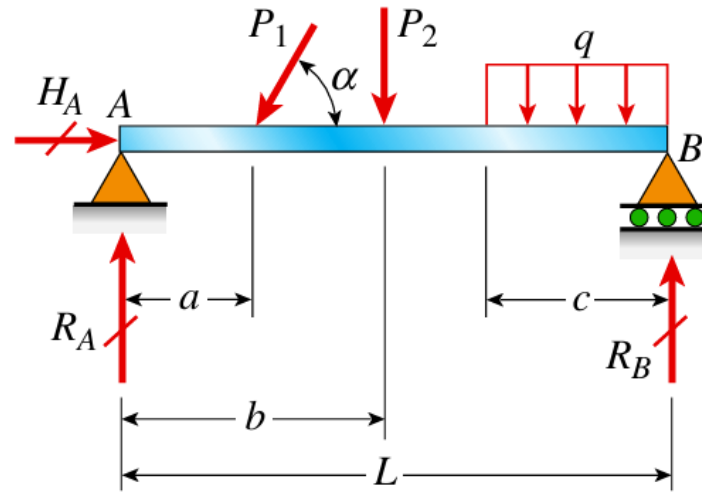
(d) Moment release ($M = 0$; $N, V \neq 0$) ★★★

Beam types



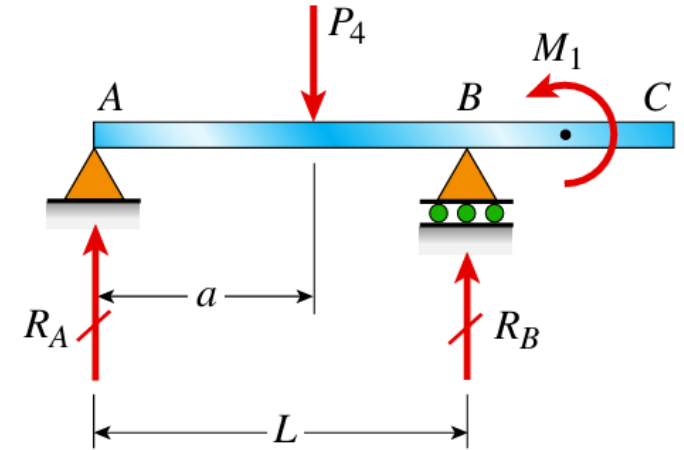
Cantilever beam

A beam which is fixed at one end and free at the other



Simple (supported) beam

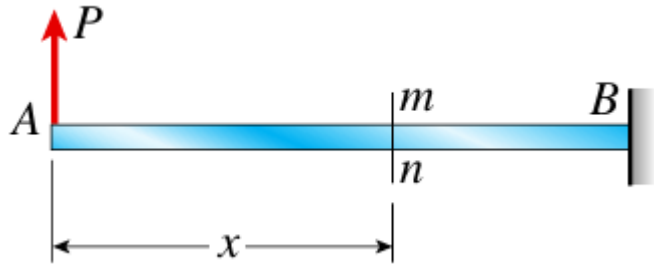
A beam with a pin support at one end and a roller support at the other



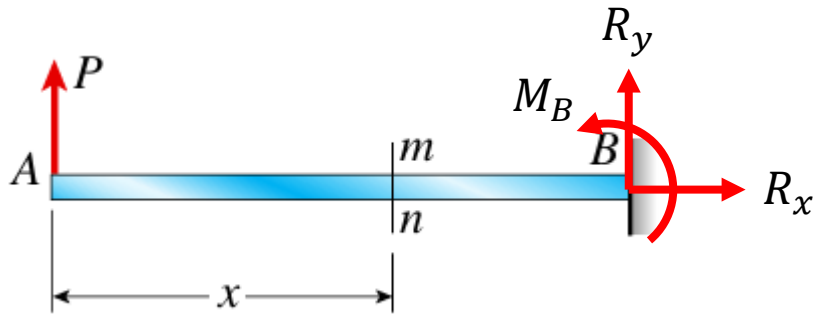
Overhang beam

Simple beam with overhang segment

Equilibrium equation of a cantilever beam



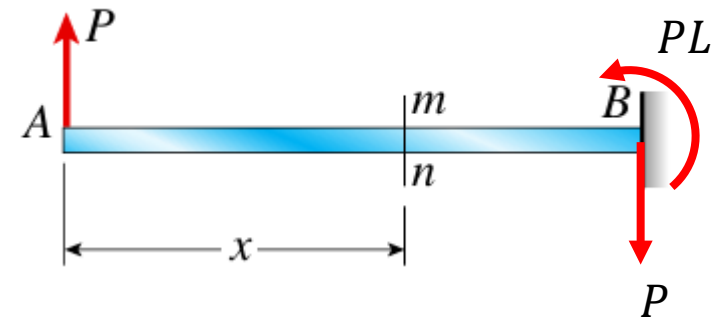
Equilibrium equation of a cantilever beam



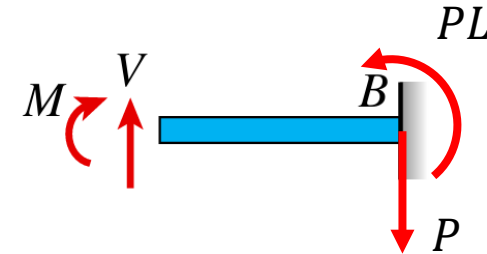
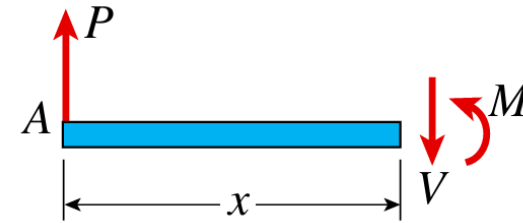
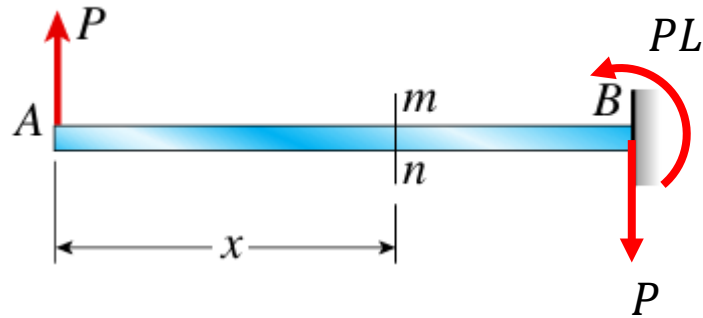
$$\begin{aligned}\sum F_x &= 0 \Rightarrow R_x = 0 \\ \sum F_y &= 0 \Rightarrow P + R_y = 0 \\ \sum M_z &= 0 \Rightarrow -P \times L + M_B = 0\end{aligned}$$



$$\begin{aligned}R_x &= 0 \\ R_y &= -P \\ M_B &= PL\end{aligned}$$



Shear forces and bending moments



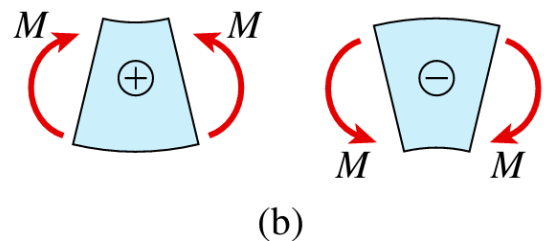
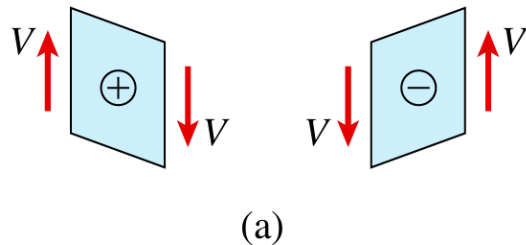
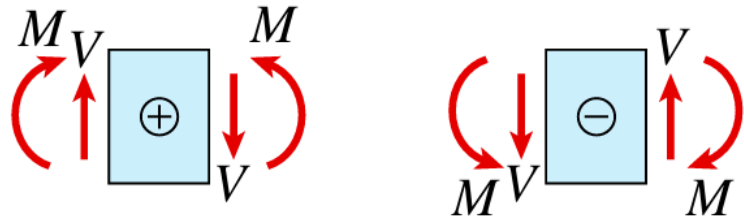
Method of sections

1. Set imaginary cuts
2. Draw FBD
3. Apply balance equation in global coordinates to get internal force V and M

$$\begin{aligned}\Sigma F_{\text{vert}} &= 0 & P - V &= 0 & \text{or} & V = P \\ \Sigma M &= 0 & M - Px &= 0 & \text{or} & M = Px\end{aligned}$$

Sign conventions

Sign convention for shear force V and bending moment M



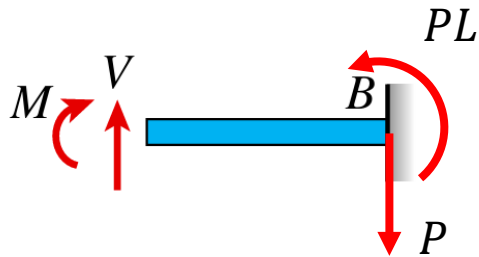
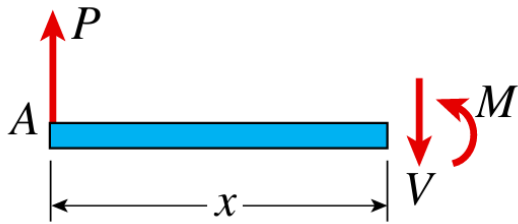
- A **positive** *shear force* acts **clockwise** against the material
- A **positive** *bending moment* **compresses the upper part** of the beam

These sign conventions are not dictated by physical necessity; they are entirely choices made by people for the sake of convenient memorization and drawing.

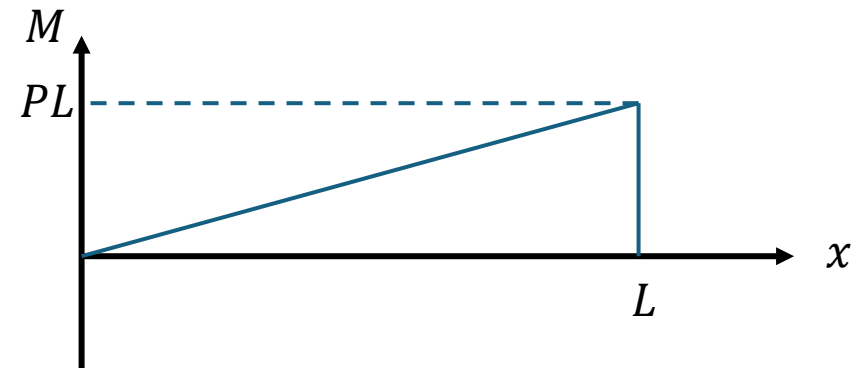
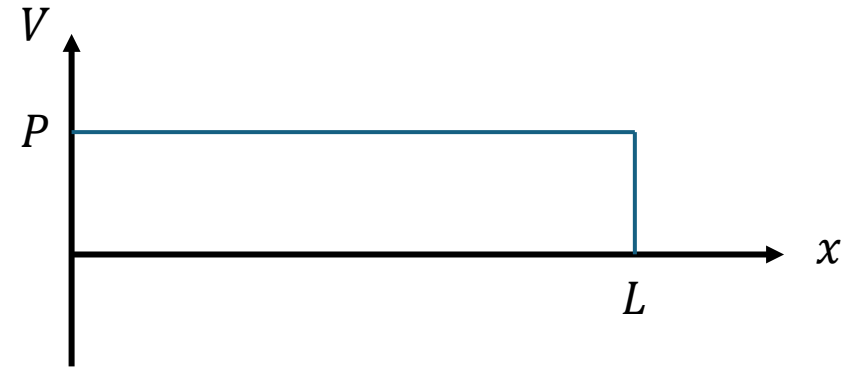
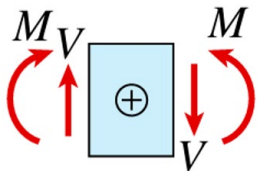


Степан Прокофьевич Тимошенко
Stepan Prokopovich Timoshenko

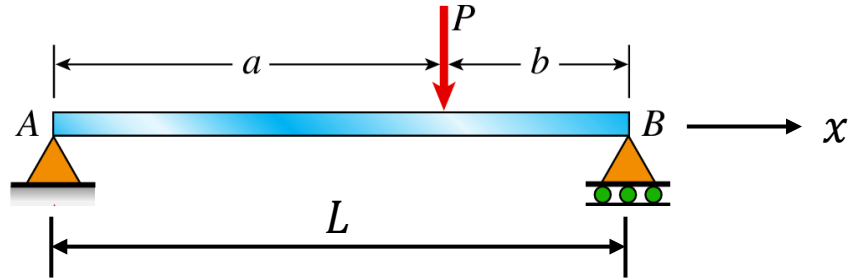
Shear-force and bending-moment diagrams



$$\begin{aligned}\Sigma F_{\text{vert}} &= 0 & P - V &= 0 & \text{or } V &= P \\ \Sigma M &= 0 & M - Px &= 0 & \text{or } M &= Px\end{aligned}$$

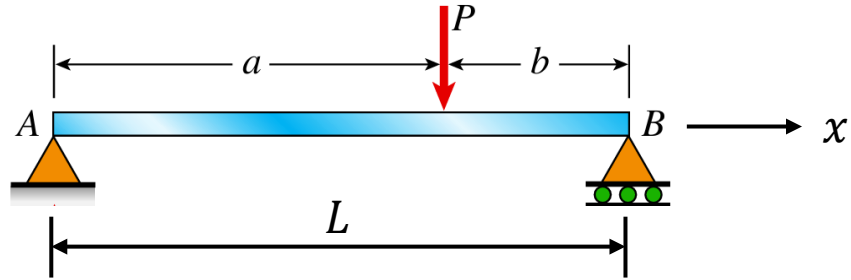


Example: simple beam + concentrated load



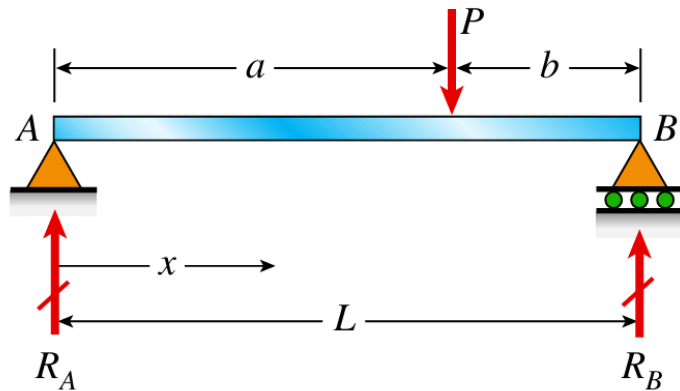
Given: P ,
draw the shear-force and bending-moment
diagrams

Example



Given: P ,
draw the shear-force and bending-moment
diagrams

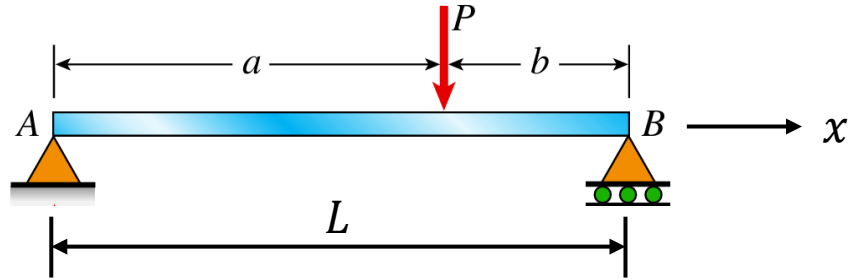
1. Draw FBD



Example

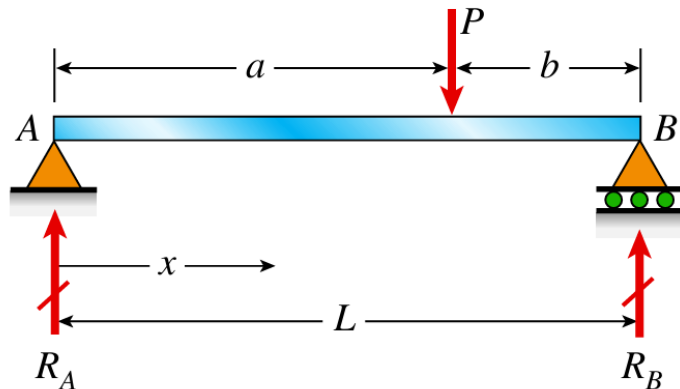
2. Equilibrium equations for reactions

$$\begin{aligned} R_A + R_B &= P \\ -Pa + R_B L &= 0 \end{aligned} \quad \Rightarrow \quad R_A = \frac{Pb}{L} \quad R_B = \frac{Pa}{L}$$

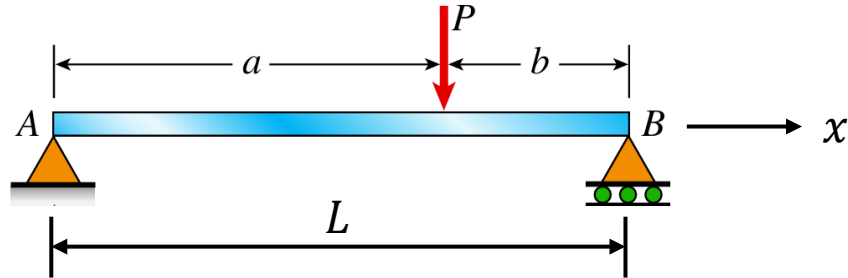


Given: P ,
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diagrams

1. Draw FBD

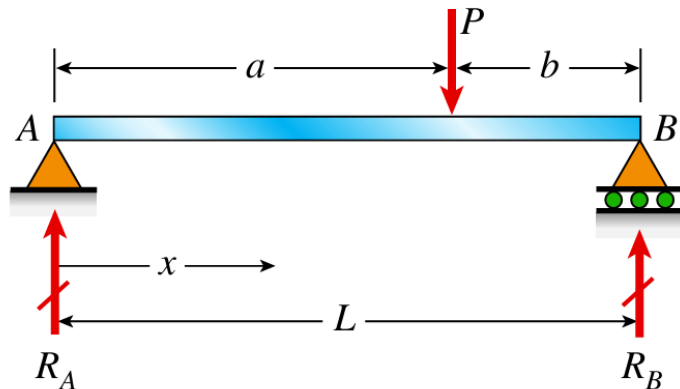


Example



Given: P ,
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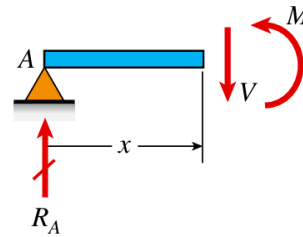
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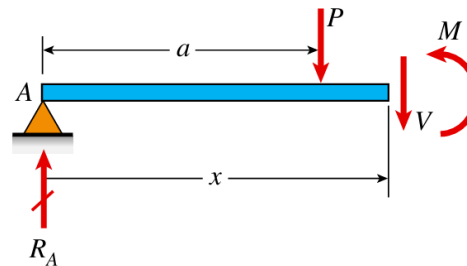
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3. Make cuts, and apply equilibrium equations



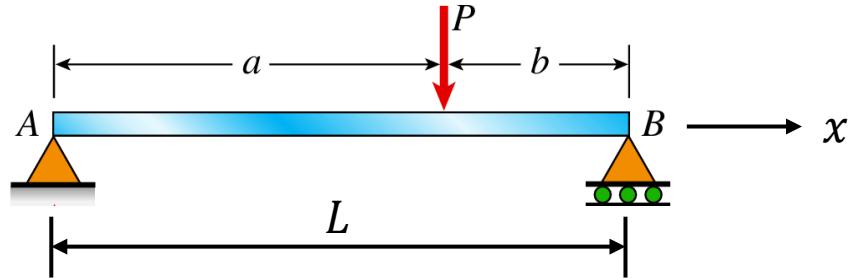
$$V = R_A = \frac{Pb}{L} \quad M = R_A x = \frac{Pbx}{L} \quad (0 < x < a)$$



$$V = R_A - P = \frac{Pb}{L} - P = -\frac{Pa}{L} \quad (a < x < L)$$

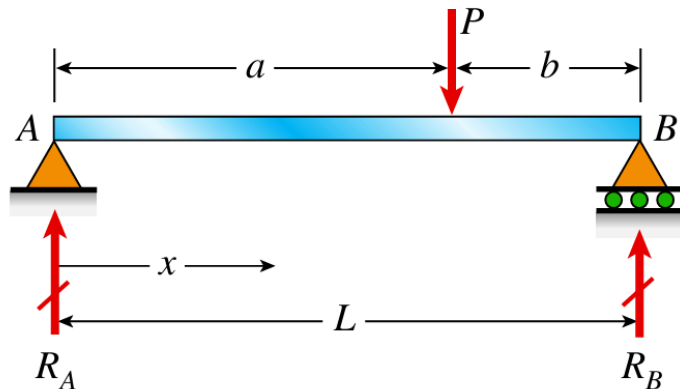
$$\begin{aligned} M &= R_A x - P(x - a) = \frac{Pbx}{L} - P(x - a) \\ &= \frac{Pa}{L}(L - x) \quad (a < x < L) \end{aligned}$$

Example



Given: P ,
draw the shear-force and bending-moment
diagrams

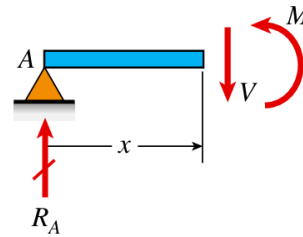
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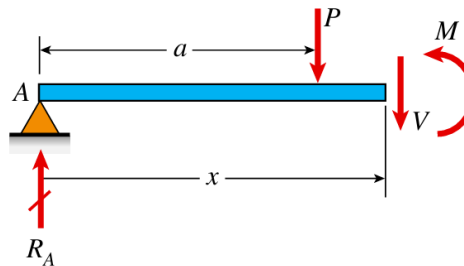
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3. Make cuts, and apply equilibrium equations



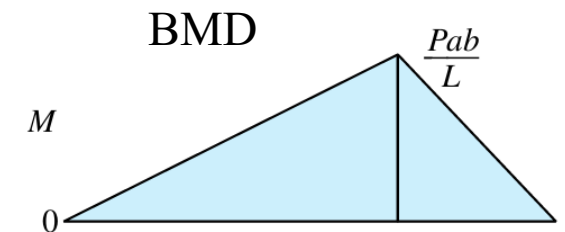
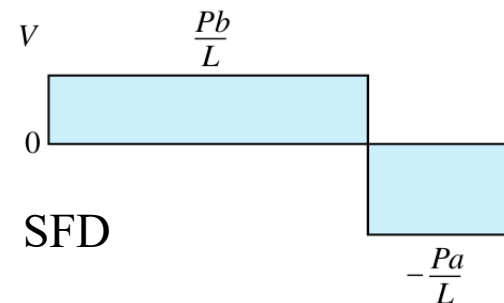
$$V = R_A = \frac{Pb}{L} \quad M = R_A x = \frac{Pbx}{L} \quad (0 < x < a)$$



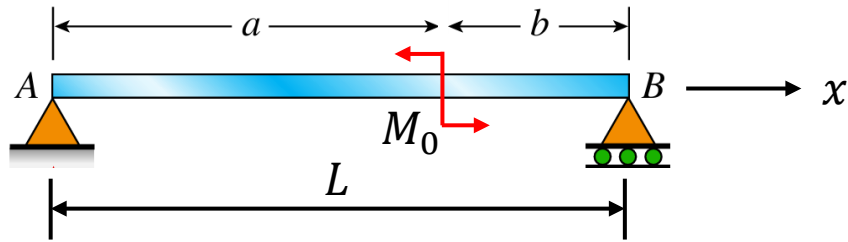
$$V = R_A - P = \frac{Pb}{L} - P = -\frac{Pa}{L} \quad (a < x < L)$$

$$\begin{aligned} M &= R_A x - P(x - a) = \frac{Pbx}{L} - P(x - a) \\ &= \frac{Pa}{L}(L - x) \quad (a < x < L) \end{aligned}$$

4. Plot

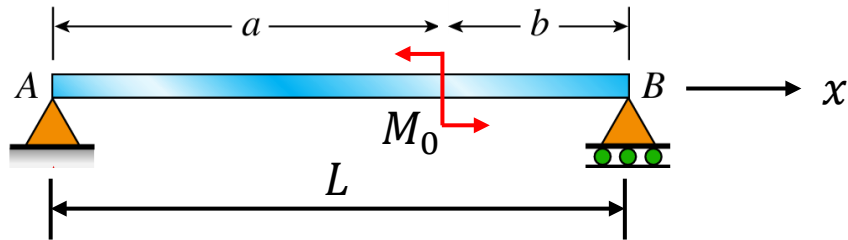


Example: simple beam + concentrated moment

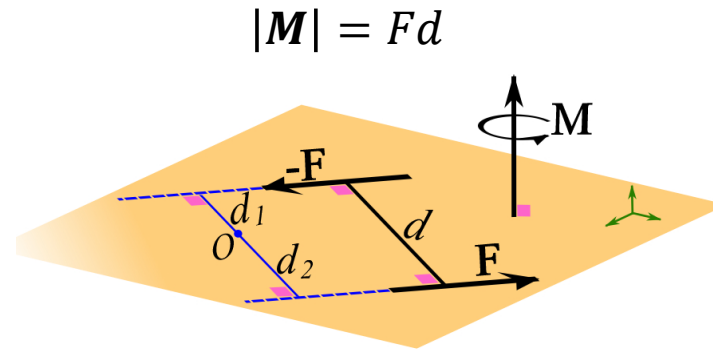
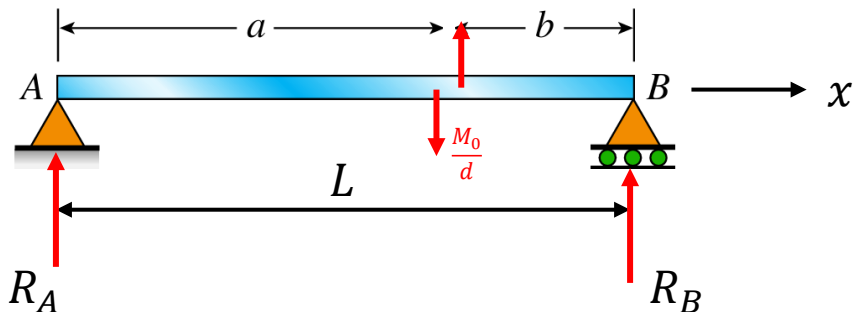


Given: M ,
draw the shear-force and bending-moment
diagrams

Example



Given: M ,
draw the shear-force and bending-moment
diagrams



A concentrated moment produces, **at any point in its plane** of action, a moment equal in magnitude to itself.

$$M_0 + R_B L = 0$$

$$M_0 - R_A L = 0$$

$$R_A + R_B + \frac{M_0}{d} - \frac{M_0}{d} = 0$$

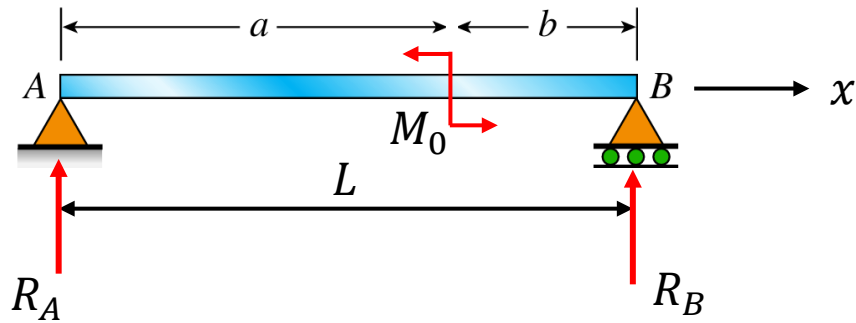
$$-\frac{M_0}{d} \left(a - \frac{d}{2} \right) + \frac{M_0}{d} \left(a + \frac{d}{2} \right) + R_B L = 0$$



$$R_A = \frac{M_0}{L}, \quad R_B = -\frac{M_0}{L}$$

Example

1. Draw FBD



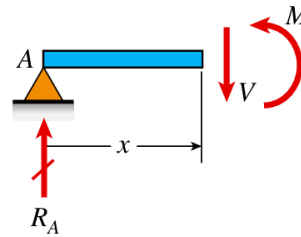
2. Equilibrium equations for reactions

$$R_A + R_B = 0$$

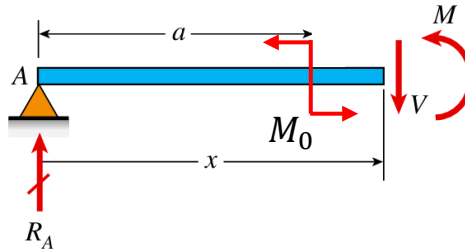
$$M_0 + R_B L = 0$$

$$\Rightarrow R_A = \frac{M_0}{L}, \quad R_B = -\frac{M_0}{L}$$

3. Make cuts, and apply equilibrium equations



$$V = R_A = \frac{M_0}{L}, \quad M = R_A x = \frac{M_0 x}{L} \quad (0 < x < a)$$

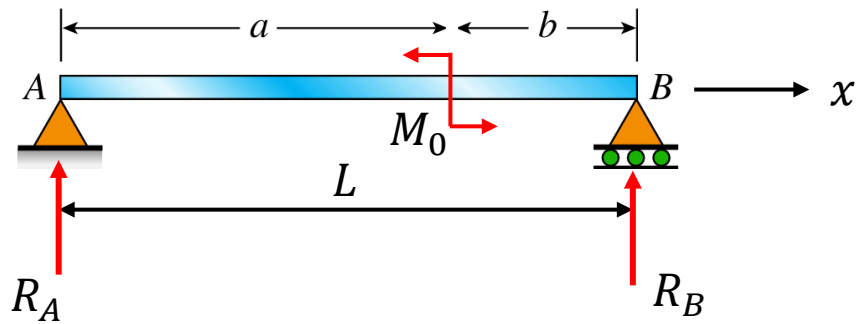


$$V = R_A = \frac{M_0}{L}$$

$$M = R_A x - M_0 = \frac{M_0}{L} (x - L) \quad (a < x < L)$$

Example

1. Draw FBD



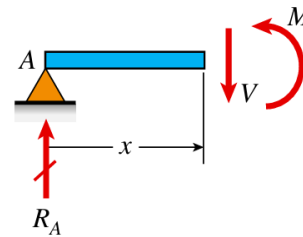
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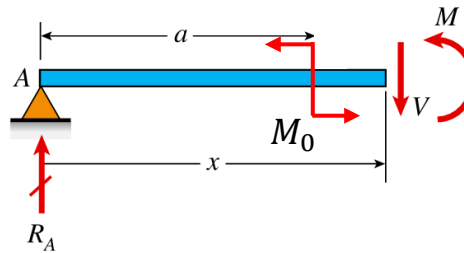
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3. Make cuts, and apply equilibrium equations



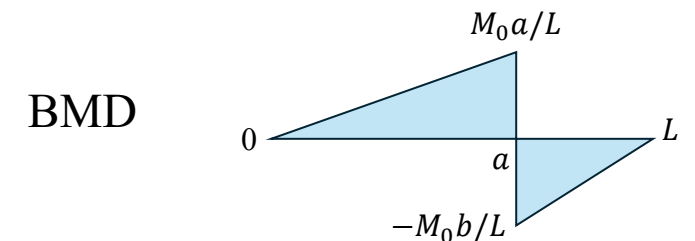
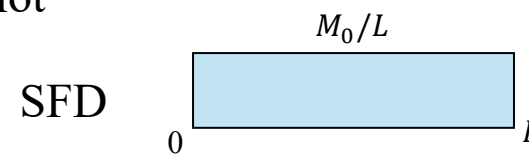
$$V = R_A = \frac{M_0}{L}, \quad M = R_A x = \frac{M_0 x}{L} \quad (0 < x < a)$$



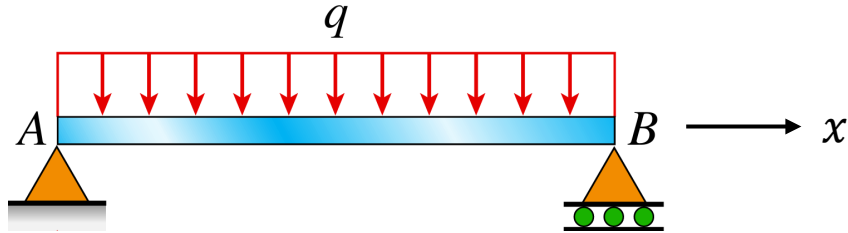
$$V = R_A = \frac{M_0}{L}$$

$$M = R_A x - M_0 = \frac{M_0}{L} (x - L) \quad (a < x < L)$$

4. Plot

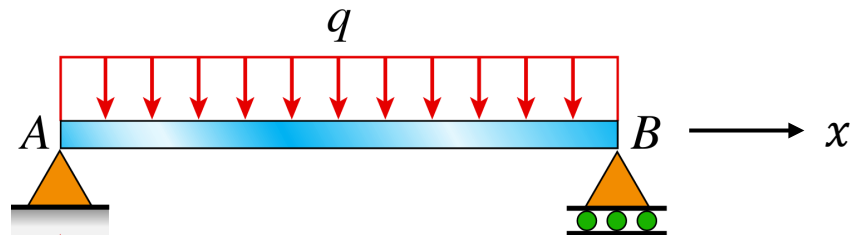


Example: simple beam + uniform load



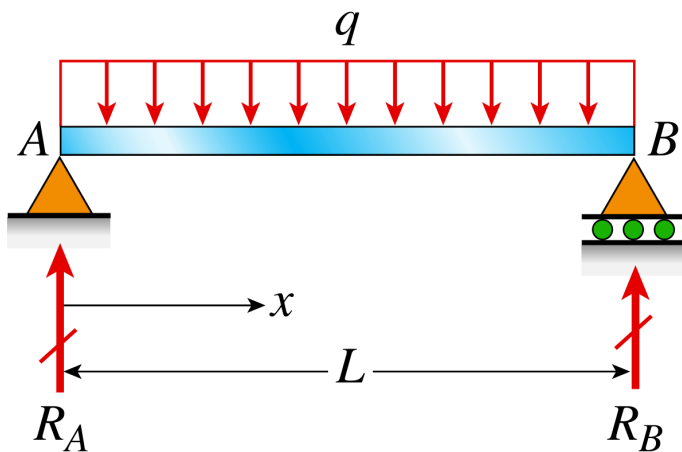
Given: q ,
draw the shear-force and bending-moment
diagrams

Example



Given: q ,
draw the shear-force and bending-moment
diagrams

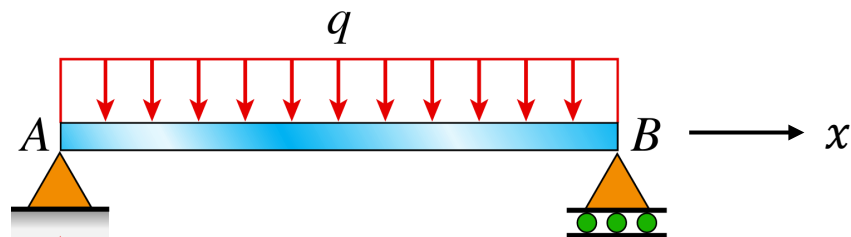
1. Draw FBD



2. Equilibrium equations for reactions

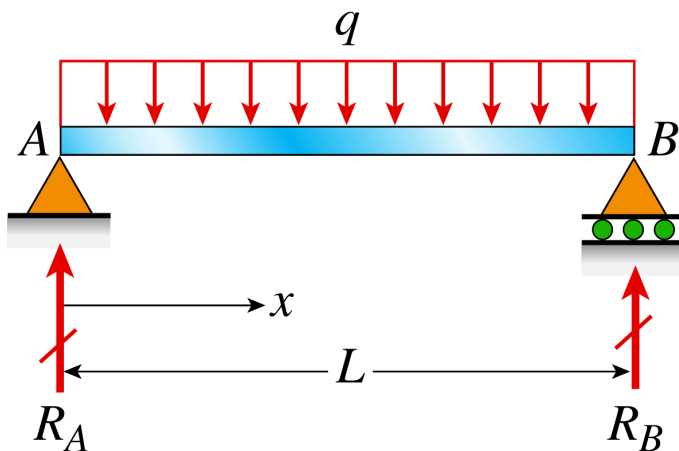
$$\begin{aligned} R_A + R_B &= qL \\ -qL \times \frac{L}{2} + R_B L &= 0 \end{aligned} \quad \Rightarrow \quad \begin{aligned} R_A &= \frac{1}{2} qL \\ R_B &= \frac{1}{2} qL \end{aligned}$$

Example



Given: q ,
draw the shear-force and bending-moment
diagrams

1. Draw FBD



2. Equilibrium equations for reactions

$$R_A + R_B = qL$$

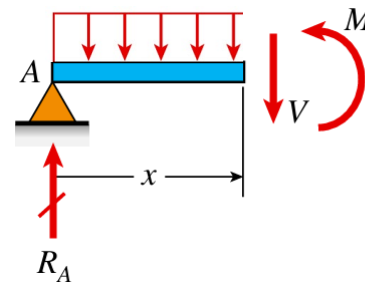
$$-qL \times \frac{L}{2} + R_B L = 0$$

➔

$$R_A = \frac{1}{2}qL$$

$$R_B = \frac{1}{2}qL$$

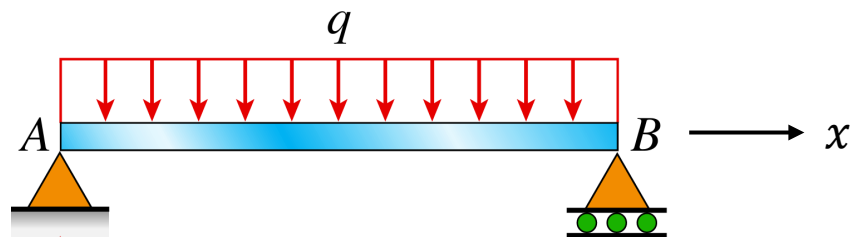
3. Make cuts, and apply equilibrium equations



$$V = R_A - qx = \frac{qL}{2} - qx$$

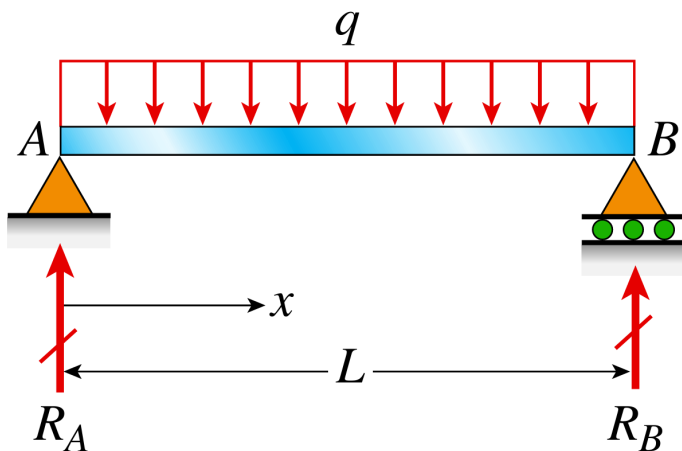
$$M = R_A x - qx \left(\frac{x}{2} \right) = \frac{qLx}{2} - \frac{qx^2}{2}$$

Example



Given: q ,
draw the shear-force and bending-moment
diagrams

1. Draw FBD



2. Equilibrium equations for reactions

$$R_A + R_B = qL$$

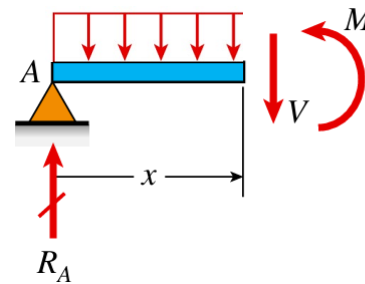
$$-qL \times \frac{L}{2} + R_B L = 0$$

➔

$$R_A = \frac{1}{2}qL$$

$$R_B = \frac{1}{2}qL$$

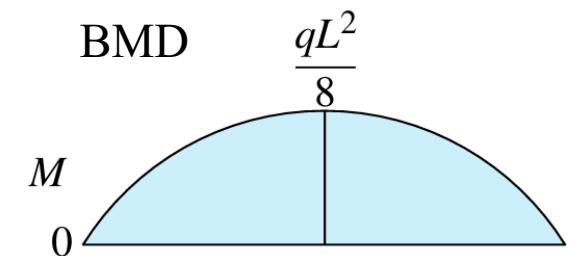
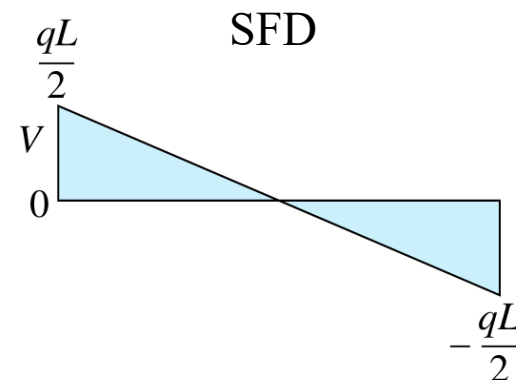
3. Make cuts, and apply equilibrium equations



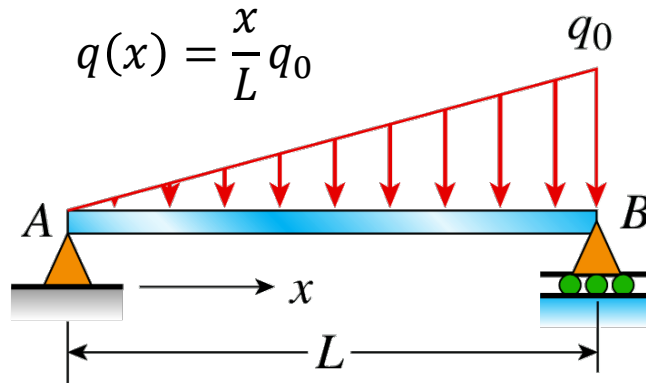
$$V = R_A - qx = \frac{qL}{2} - qx$$

$$M = R_A x - qx \left(\frac{x}{2} \right) = \frac{qLx}{2} - \frac{qx^2}{2}$$

4. Plot



Exercise



Hint:

A linearly distributed triangular load is equivalent to a single concentrated force of equal magnitude applied at the centroid, located at the last one-third of the span (i.e., measured from the larger-intensity end of the load).

Assume $q(x) = \frac{q_0}{L} x$.

1. Resultant (equivalent point load)

$$F = \int_0^L q(x) dx = \int_0^L \frac{q_0}{L} x dx = \frac{q_0}{2} L.$$

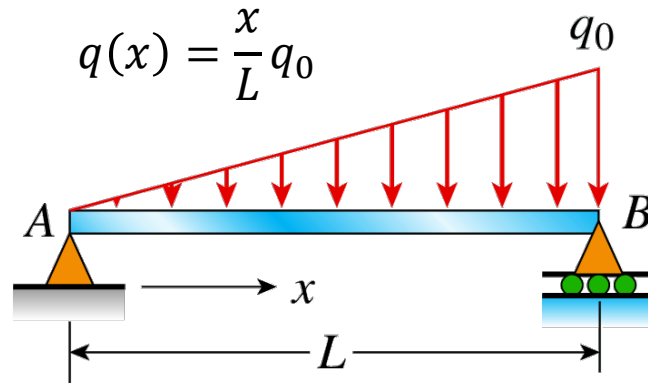
2. Location (from the left end)

$$F \bar{x} = \int_0^L x q(x) dx = \int_0^L x \cdot \frac{q_0}{L} x dx = \frac{q_0}{L} \int_0^L x^2 dx = \frac{q_0}{L} \cdot \frac{L^3}{3} = \frac{q_0 L^2}{3}.$$

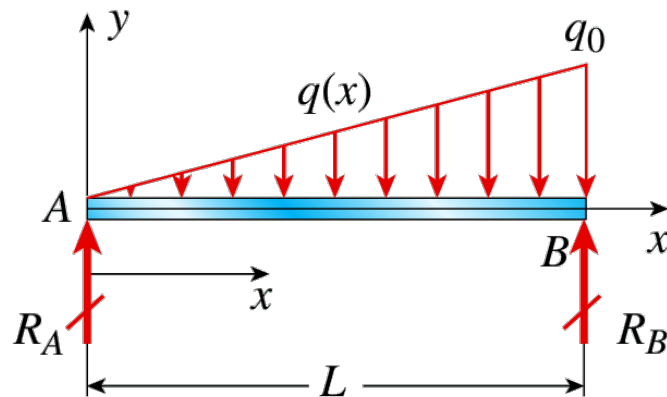
Hence

$$\bar{x} = \frac{\frac{q_0 L^2}{3}}{\frac{q_0}{2} L} = \frac{2L}{3}.$$

Exercise



1. Draw FBD



2. Equilibrium equations for reactions

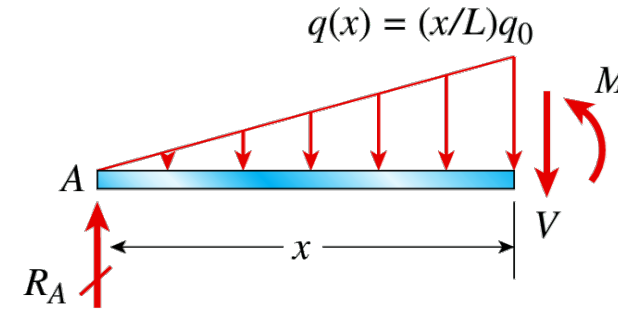
$$\Sigma F_y = 0 \quad R_A + R_B - \frac{1}{2}q_0L = 0$$

$$\Sigma M_A = 0 \quad R_B L - \frac{1}{2}(q_0L)\left(\frac{2}{3}L\right) = 0$$

$$R_A = -R_B + \frac{1}{2}q_0L = \frac{1}{6}q_0L$$

$$R_B = \frac{1}{3}q_0L$$

3. Make cuts, and apply equilibrium equations



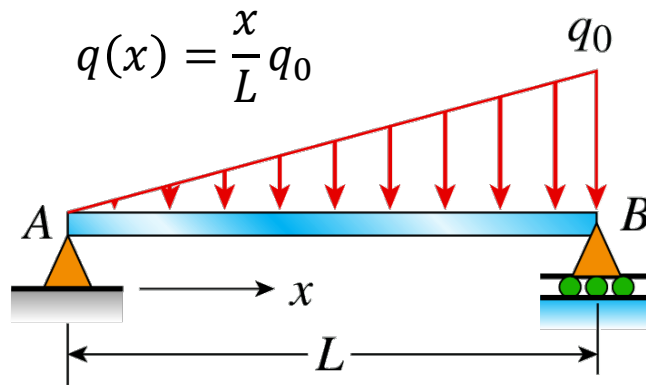
$$\Sigma F_y = 0 \quad V(x) = R_A - \frac{1}{2}\left(\frac{x}{L}q_0\right)(x) = \frac{q_0L}{6} - \frac{1}{2}\left(\frac{x}{L}q_0\right)(x)$$

$$V(x) = \frac{q_0}{6L}(L^2 - 3x^2)$$

$$\Sigma M = 0 \quad M(x) = R_A(x) - \frac{1}{2}\left(\frac{x}{L}q_0\right)(x)\left(\frac{x}{3}\right) = \frac{q_0L}{6}(x) - \frac{1}{6}\left(\frac{x^3}{L}q_0\right)$$

$$M(x) = \frac{q_0x}{6L}(L^2 - x^2)$$

Exercise

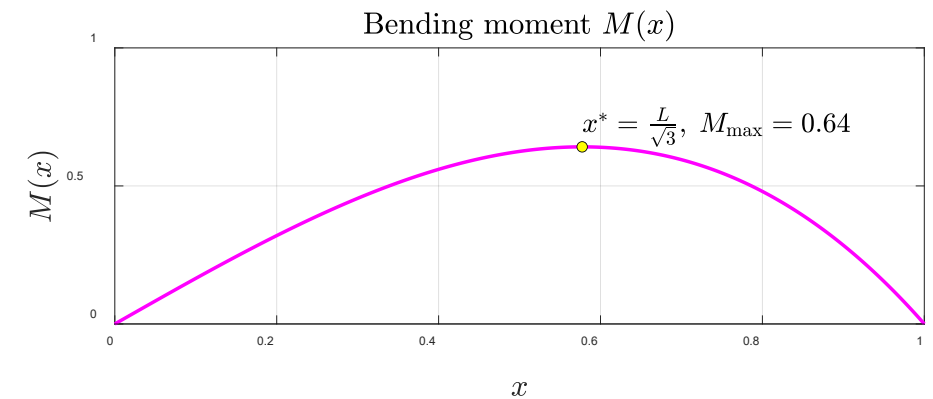
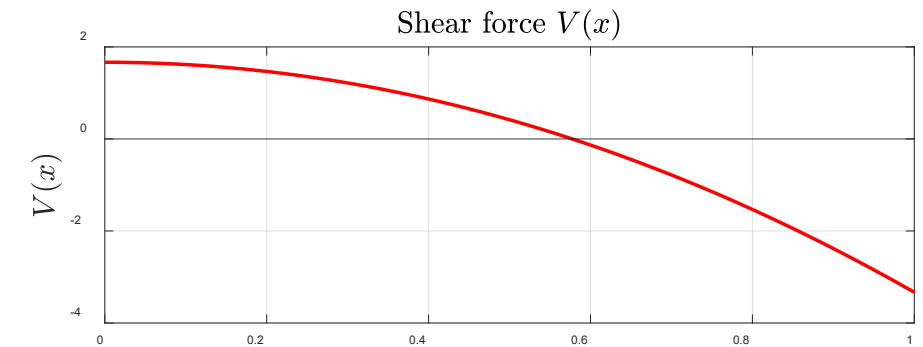
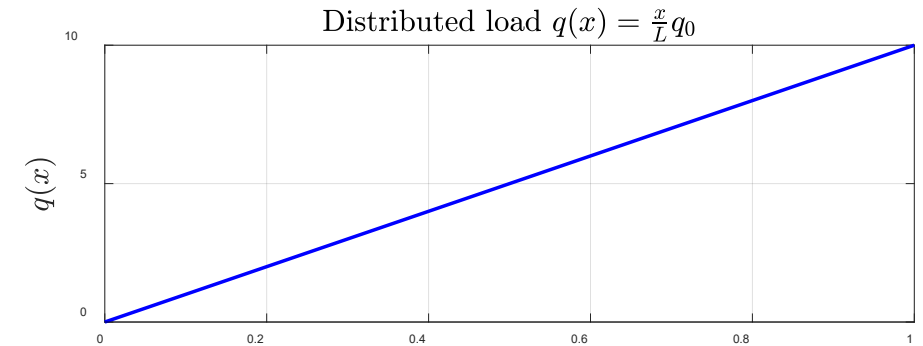


$$q(x) = \frac{x}{L}q_0$$

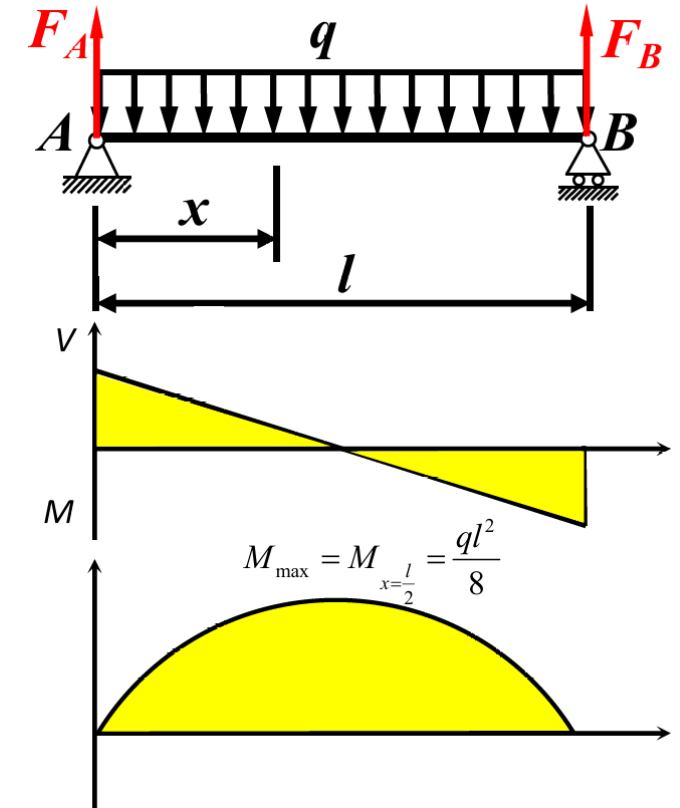
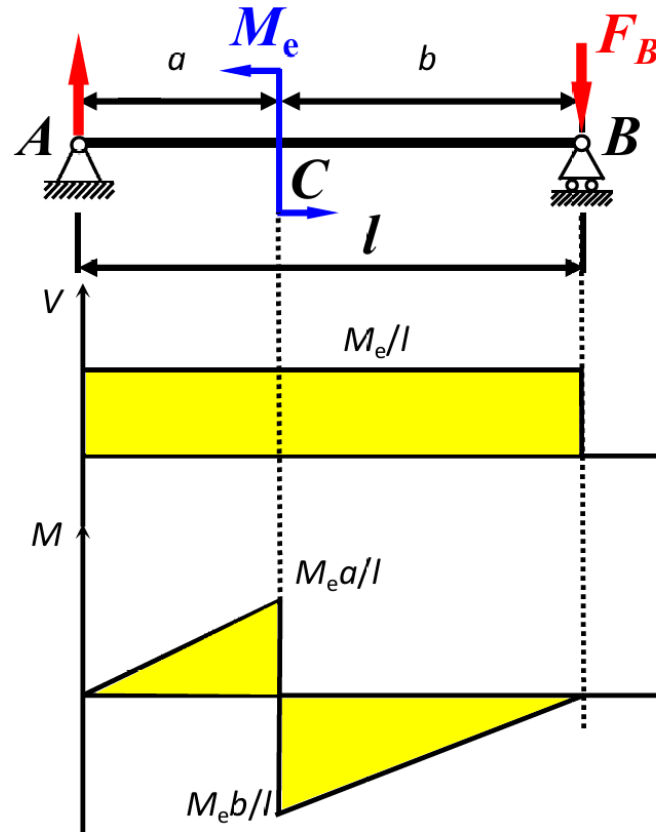
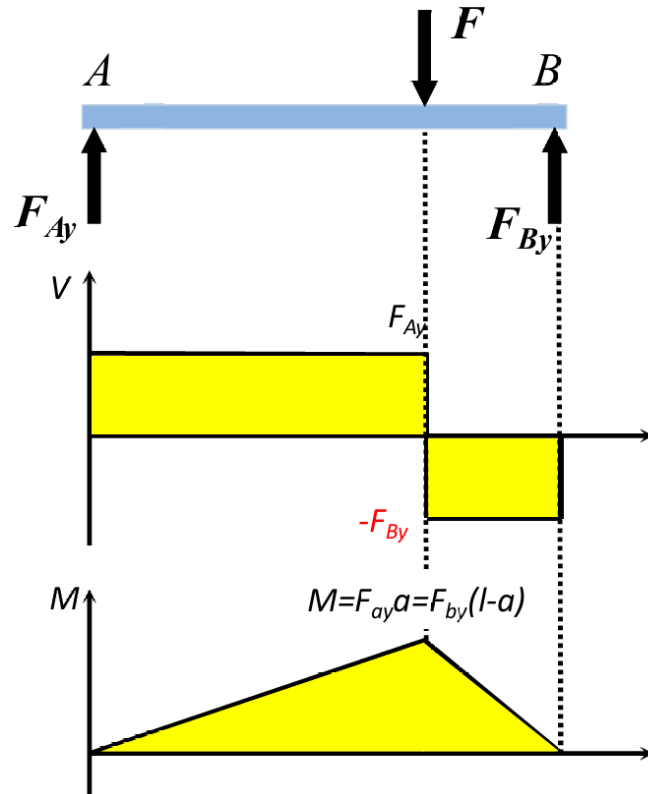
$$V(x) = \frac{q_0}{6L}(L^2 - 3x^2)$$

If $q_0 = 10 \text{ N/m}$, $L = 1 \text{ m}$, then we can plot the SFD and BMD

$$M(x) = \frac{q_0 x}{6L}(L^2 - x^2)$$

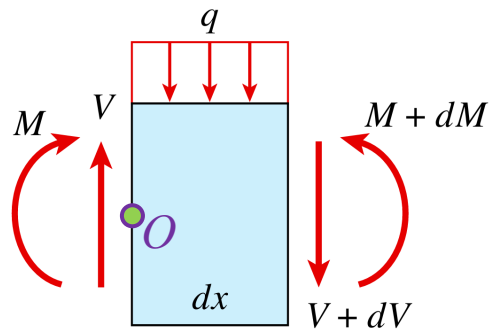


Relationships among $q(x)$, $V(x)$, and $M(x)$



Relationships among $q(x)$, $V(x)$, and $M(x)$

$$\Sigma F_{\text{vert}} = 0 \quad V - q dx - (V + dV) = 0$$



$$\Rightarrow \frac{dV}{dx} = -q \Rightarrow V_B - V_A = -\int_A^B q dx$$

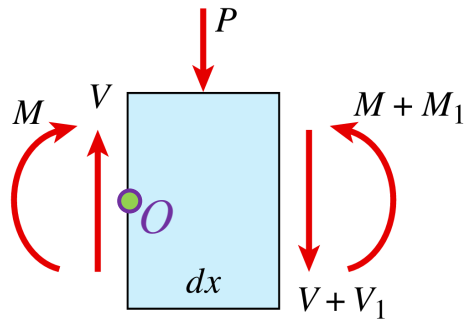
$= -(\text{area of the loading diagram between } A \text{ and } B)$

$$\Sigma M = 0 \quad -M - q dx \left(\frac{dx}{2} \right) - (V + dV) dx + M + dM = 0$$

$$\Rightarrow \frac{dM}{dx} = V \Rightarrow M_B - M_A = \int_A^B V dx$$

$= (\text{area of the shear-force diagram between } A \text{ and } B)$

Jumps due to the concentrated load



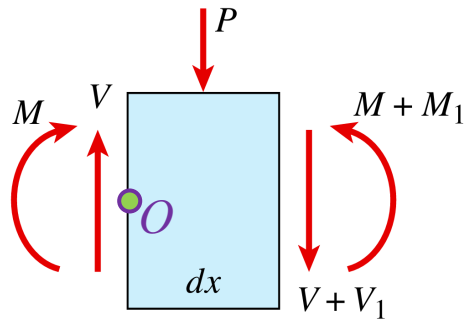
$$\Sigma F_{\text{vert}} = 0 \quad V - P - (V + V_1) = 0 \quad \Rightarrow \quad V_1 = -P$$

$$\Sigma M = 0 \quad -M - P\left(\frac{dx}{2}\right) - (V + V_1)dx + M + M_1 = 0$$

$$\Rightarrow \quad M_1 = P\left(\frac{dx}{2}\right) + Vdx + V_1dx$$

Since the length dx of the element is infinitesimally small, the increment M_1 in the bending moment is also infinitesimally small. Thus, **the bending moment does not change when passing through the point of application of a concentrated load.**

Jumps due to the concentrated load

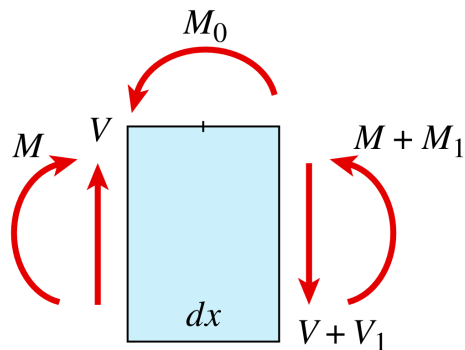


$$\Sigma F_{\text{vert}} = 0 \quad V - P - (V + V_1) = 0 \quad \Rightarrow \quad V_1 = -P$$

$$\Sigma M = 0 \quad -M - P\left(\frac{dx}{2}\right) - (V + V_1)dx + M + M_1 = 0$$

$$\Rightarrow \quad M_1 = P\left(\frac{dx}{2}\right) + Vdx + V_1dx$$

Since the length dx of the element is infinitesimally small, the increment M_1 in the bending moment is also infinitesimally small. Thus, **the bending moment does not change when passing through the point of application of a concentrated load.**



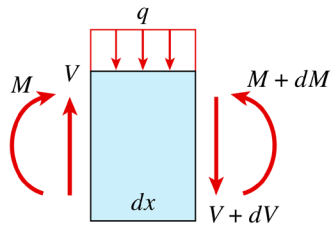
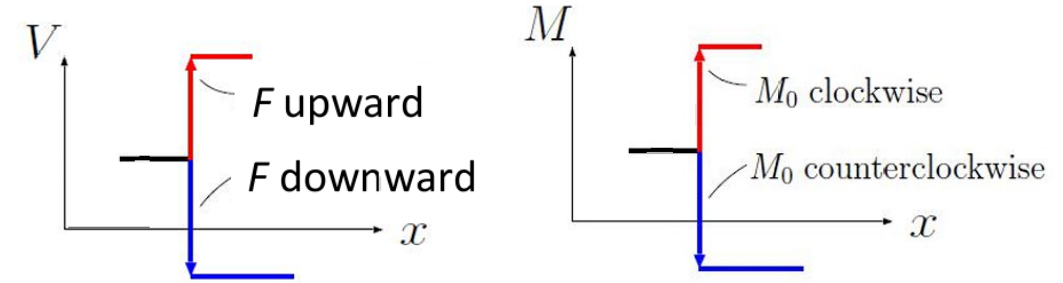
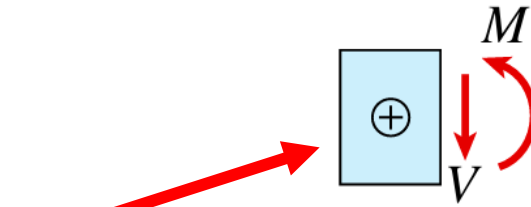
$$\Sigma M = 0 \quad -M + M_0 - (V + V_1)dx + M + M_1 = 0$$

$$\Rightarrow \quad M_1 = -M_0$$

This relation shows that the bending moment decreases by M_0 when moving from left to right through the point of load application. Thus, **the bending moment changes abruptly at the point of application of a couple/moment.**

Summary

From **left to right !!!!!**



$$\frac{dV}{dx} = -q(x) \quad q(x) \text{ downward}$$

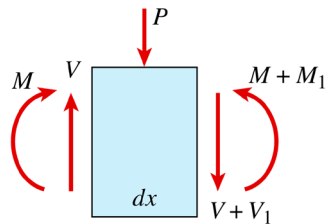


Upward $q(x) \rightarrow V(x)$ increase
Integrate $q(x)$ to get change in $V(x)$

$$\frac{dM}{dx} = V(x)$$



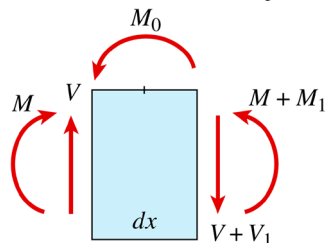
Integrate $V(x)$ to get change in $M(x)$



$$\Delta V = -P \quad P \text{ downward}$$



Upward $P \rightarrow V(x)$ upward jump

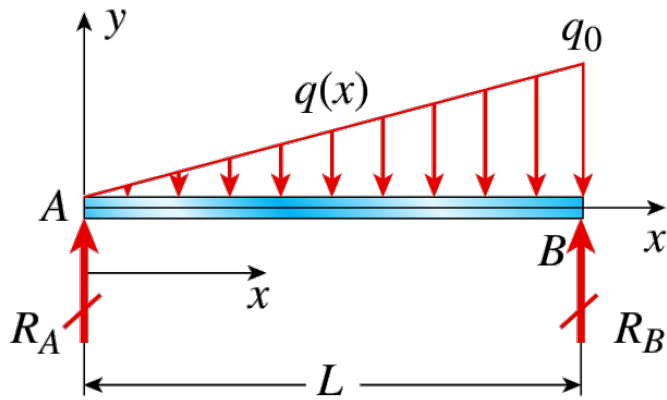


$$\Delta M = -M_0 \quad M_0 \text{ counterclockwise}$$



Clockwise $M_0 \rightarrow M(x)$ upward jump

Recall



$$R_A = -R_B + \frac{1}{2}q_0L = \frac{1}{6}q_0L$$

$$R_B = \frac{1}{3}q_0L$$

$$q(x) = \frac{x}{L}q_0$$

$$\begin{aligned} V(x) &= V_A + \int_A^x -q(x)dx = R_A + \int_0^x -\frac{x}{L}q_0dx \\ &= \frac{1}{6}q_0L - \frac{q_0}{2L}x^2 \Big|_0^x = \frac{1}{6}q_0L - \frac{q_0}{2L}x^2 = \frac{q_0}{6L}(L^2 - 3x^2) \end{aligned}$$

$$\begin{aligned} M(x) &= M_A + \int_A^x V(x)dx = 0 + \int_0^x \frac{q_0}{6L}(L^2 - 3x^2)dx \\ &= \frac{q_0}{6L}(L^2x - x^3) \Big|_0^x = \frac{q_0x}{6L}(L^2 - x^2) \end{aligned}$$